How can we change colour? (1)

Aim

To investigate the effect of coloured filters first on white light, and then on the spectrum of white light produced by a prism.

Equipment

- ray box with single slit and power supply
- red, blue and green filters
- plain paper
- equilateral triangular glass prisms



Predict (1)

❶ Predict what you expect to see when:

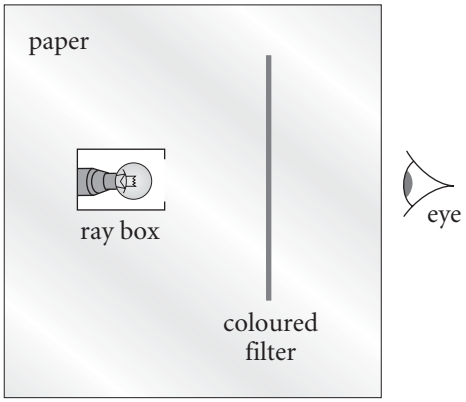
a a red filter is placed in front of the ray lamp

b a blue filter is placed in front of the ray lamp

c a green filter is placed in front of the ray lamp

d a blue and a green filter are placed together in front of the ray lamp.

What to do



❷ Switch on the ray box (without the slit) to give a beam of white light.

❸ Place a red filter in front of the ray lamp. What do you see?

❹ Repeat step ❸ using blue and green filters in turn.

❺ Repeat step ❸ using the blue and green filters together.

Questions

1 Were your predictions correct?

2 What do the filters do to the white light?

Continued ⟶

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How can we change colour? (1) continued

Predict (2)

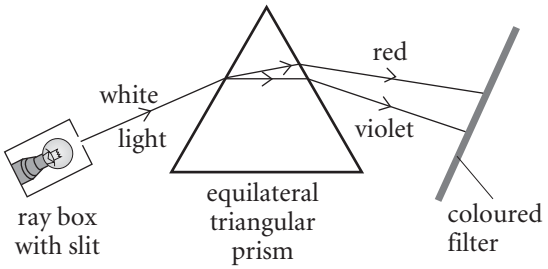
❻ Predict what you expect to see when:

a a red filter is placed in the white light spectrum produced by a prism

b a blue filter is placed in the white light spectrum produced by a prism

c a green filter is placed in the white light spectrum produced by a prism.

What to do now





❼ Use a triangular prism and a ray box with a single slit to produce a spectrum of white light.

❽ Place a red filter in the path of the spectrum. What do you see?

❾ Repeat step ❽ using blue and green filters in turn.

Questions

3 Were your predictions correct?

4 What happens to the light that does not pass through the filters?

5 How could you use a second prism to recombine the colours to make white light again? Try it out.

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Investigating sounds

L1a
Core

Aim

To investigate a number of different ways in which sounds are produced.

Equipment

Activity A

- tuning forks of different frequencies
- large rubber bung
- ping-pong ball hanging on a thread
- clamp stand

Activity B

- board with six nails and elastic bands stretched between them

What to do

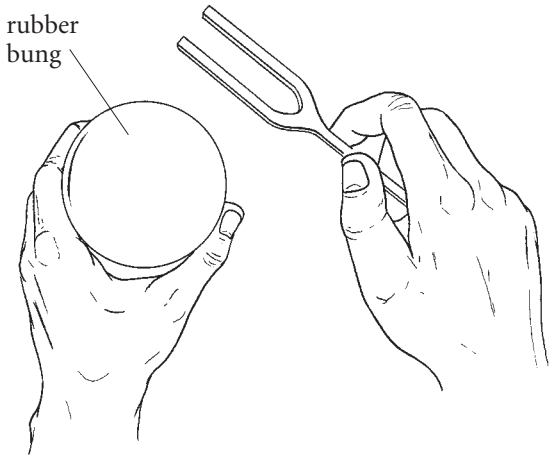
Carry out the experiments below, and for each one answer these three questions.

1 What is producing the sound?

2 How can the sound be made louder or softer?

3 How can the sound be made higher pitched or lower pitched?

A The tuning fork



❶ Hold the tuning fork by the stem and hit one prong firmly against the rubber bung. Look closely at the fork. What is happening while it is making the sound?

❷ Sound the tuning fork again and touch one prong against the hanging ping-pong ball. What happens to the ball?

❸ Hit the rubber bung softly with the fork, so it just makes a sound, and then touch the ping-pong ball again. Is there any difference between hitting the tuning fork firmly and softly?

❹ Sound a different sized tuning fork. Is the note higher or lower than with the first tuning fork?

B The rubber band



❺ Wearing goggles, pluck one of the rubber bands sideways and look closely at the band while it is making the sound. What does it do?

❻ Pluck the other rubber bands. What difference do you notice in the sound? Can you see a pattern? (The longer the band, the . . .)

❼ Try plucking the same rubber band softly and then firmly. Is there a difference in the sound?

Continued ⟶

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